Battery

Component	Errorcode	Short Text	Detection	Description	Action
Battery:BMWix	0x1001	OneContactorSticky	**BMWix:StateErrorContactor** == OneContactorSticky	SME detected than one contactor is sticky. This only plausibilized but not directly measured. This error normally occur, when system is on through shorepower and plug is pulled out. When this error is active the DCU will erase the failure memory of the battery	Due to the automatic erase of that error it should normally not be active, but if so read failure memory with Tool32
Battery:BMWix	0x1002	TwoContactorsSticky	**BMWix:StateErrorContactor** == TwoContactorsSticky	SME detected than both contactors are sticky. This only plausibilized but not directly measured.	Read failure memory with Tool32
Battery:BMWix	0x1003	PrechargeLock	**BMWix:StatePrechargeLock** == Active	SME inhibit Precharge process, due to other errors	Check for short circuit or too high capacitance on HV bus. Check for other battery errors
Battery:BMWix	0x3002	SocLow	10.0% < **BMWix:GStateOfCharge** (UserSOC) <= 20.0%	Battery SOC low	charge battery
Battery:BMWix	0x3003	SocVeryLow	1.0% < **BMWix:GStateOfCharge** (UserSOC) <= 10.0%	Battery SOC low	charge battery
Battery:BMWix	0x3004	SocCriticalLow	**BMWix:GStateOfCharge** (UserSOC) <= 1.0%	Battery SOC low, or no CAN communication to battery	charge battery, check connection (e.g. **PackVoltage** in component details)
Battery:BMWix	0x6001	HVIL_Fail	**BMWix:StateErrorLock** == Active	HV Interlock line not connected	Check HV interlock plug above battery HV plug
Battery:BMWix	0x7001	Communication_Error	No CAN message received for min. 1 sec.	No CAN communication to battery	Check connection to battery (CAN and supply)
Battery:BMWix	0x9001	CellVoltageDeviation	**BMWix:MaxCellVoltage** > 0.0V and **BMWix:MinCellVoltage** > 0.0V and **Battery:GCurrent** < 30.0A and Difference between Max- and Min Cell Voltage > 40.0mV	Battery cells are disbalanced and need balancing	Switch off the system (battery contactors == open and Highside == low) but keep continous 12V applied (do not disconnect starter battery)
Battery:BMWix	0xA001	TempHigh	SE07, SE09 **BMWix:ActualTempBatteryMax** >= 41°C SP15 **BMWix:ActualTempBatteryMax** == 48°C	SE07, SE09 Highest measured temperature inside battery is high. Error will reset if temperature is below 39°C SP15 Highest measured temperature inside battery is 48°C.	Enable cooling, reduce battery current
Battery:BMWix	0xA003	CoolingCompConnLost	No LIN bus message from compressor received for min. 200ms	No communication to cooling compressor	Check connection to compressor
Battery:BMWix	0xA004	CoolingModuleConnLost	**BMWix:CoolingType** == Webasto and No CAN message from cooling Module received for 1.1 s	No communication to cooling module	Check cabeling and supply voltage
Battery:BMWix	0xA007	CoolingCompSdOverUnderVoltage	**BMWix:EkmvDeratingReason** == ShutdownOverUnderVoltage	(HV Voltage <= 190V or HV Voltage >= 425V) and compressor start request for min. 2s	Check HV plug and cabeling to compressor
Battery:BMWix	0xA008	CoolingCompSdCriticalOvertemp	**BMWix:EkmvDeratingReason** == ShutdownCriticalOvertemp	Temp sensor of power output stage above 125°C	Check cooling policy to be not *immediate* , Check Cooling valve on battery, Check cooling water pump
Battery:BMWix	0xA009	CoolingCompSdTorqueExceeded	**BMWix:EkmvDeratingReason** == ShutdownTorqueExceeded	Motor phase current > 40A	Check Cooling circuit, replace compressor

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Battery:BMWix	0xA00A	CoolingCompSdLinCommError	**BMWix:EkmvDeratingReason** == ShutdownLinCommError	Compressor did not receive LIN message for 2.2s	Check LIN cabeling
Battery:BMWix	0xA00B	CoolingCompSdExternalSupply	**BMWix:EkmvDeratingReason** == ShutdownExternalSupply	HV Voltage >= 425V and compressor start request for min. 2s	Check HV cabeling to compressor
Battery:BMWix	0xA00C	CoolingCompSdErrorInverterSupply	**BMWix:EkmvDeratingReason** == ShutdownErrorInverterSupply	IGBT driver voltage < 13.5V or IGBT driver voltage >= 18.5V	Replace compressor
Battery:BMWix	0xA00D	CoolingCompSdErrorSensor	**BMWix:EkmvDeratingReason** == ShutdownErrorSensor	Internal failure in compressor	Replace compressor
Battery:BMWix	0xA00E	CoolingCompSdShortcircuit	**BMWix:EkmvDeratingReason** == ShutdownShortcircuit	Motor phase current > 60A	Replace compressor
Battery:BMWix	0xA00F	CoolingCompSdRunUp	**BMWix:EkmvDeratingReason** == ShutdownRunUp	Compressor can't spin up (fail after 4 start attempts)	Check compressor, replace compressor
Battery:BMWix	0xA010	CoolingWaterFlowLow	cooling started and for 10 s after start **BMWix:EkmvFlowIn** < 100 l/min or **BMWix:EkmvFlowOut** < 100 l/min	Not enough water flow through cooling circuit	Check pump and water pipes
Battery:BMWix	0xA013	CoolingWaterTempLow	**BMWix:EkmvTempIn** < -3°C	Water possibly frozen and cooling will not start	Very unlikely, Check Temperature sensor
Battery:BMWix	0xA014	CoolingWaterTempHigh	for min. 10s **BMWix:EkmvTempIn** > 70°C	Water temperature too high for cooling to work	Check water pipes and pump operation
Battery:BMWix	0xA016	CoolingPresLow	**BMWix:EkmvPresHigh** < 0.0 bar	Under pressure in refrigerant circuit	Verify that refrigerant circuit is filled correctly
Battery:BMWix	0xA017	CoolingPresHigh	**BMWix:EkmvPresHigh** > 25.0 bar	Overpressure in refrigerant circuit, cooling will be disabled for 60s if this error occur to prevent Overtemp in compressor	Check waterflow and pump, also check Cooling valve at battery
Battery:BMWix	0xA018	CoolingWaterInTempSensor	**BMWix:EkmvTempIn** < -5°C or **BMWix:EkmvTempIn** > 100°C	Sensor error in water in sensor. This error will stop cooling	Check sensor cabeling and sensor function
Battery:BMWix	0xA019	CoolingWaterOutTempSensor	**BMWix:CoolingType** == Webasto **BMWix:EkmvTempIn** < -5°C or **BMWix:EkmvTempIn** > 100°C	Sensor error in water in sensor.	Check sensor cabeling and sensor function
Battery:BMWix	0xA022	CoolingNotDetected	At DCU startup: **Dcu:AnalogIn0** < 0.25V and **Dcu:AnalogIn1** < 0.25V and No CAN message from *Webasto* cooling module received	Neither Webasto nor Ymer cooling module detected. Automatic cooling module detection did not find cooling module.	Ymer: Disconnect HS2.1, check analog in 0 (temp) and 1 (preassure) < 0,25V, if not: check wiring and sensor Webasto: Check connection and power supply to cooling module - CAN communication to module is not possible
Battery:BMWix	0xA023	TempAboveNormal	SP15 only **ActualTempBatteryMax** == 44°C	Highest measured temperature inside battery is 44°C.	Enable cooling, reduce battery current
Battery:BMWix	0xA024	TempModerately	SP15 only	Highest measured temperature inside battery is between 45°C and 46°C.	Enable cooling, reduce battery current

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			ActualTempBatteryMax is (45°C or 46°C)		
Battery:BMWix	0xA025	TempModeratelyHigh	SP15 only **ActualTempBatteryMax** == 47°C	Highest measured temperature inside battery is 47°C.	Enable cooling, reduce battery current
Battery:BMWix	0xA026	TempVeryHigh	SP15 only **ActualTempBatteryMax** == 49°C	Highest measured temperature inside battery is 49°C.	Enable cooling, reduce battery current
Battery:BMWix	0xA027	TempCritical	SP15 only **ActualTempBatteryMax** >= 50°C	Highest measured temperature inside battery is 50°C or higher.	Enable cooling, reduce battery current
Battery:BMWix	0xB002	ServiceDisconnect	**BMWix:StateServiceDisconnect** != Connected	Service Disconnect Plug not present and Contactor wouldn't close. Not supplied in TQ System.	Due to no Service Disconnect in TQ System, look for other battery errors and check communication connection.
Battery:BMWix	0xB003	AbortCharging	**BMWix:RequestAbortCharging** == AbortRequested or ChargingNotPossible and **BMWix:ActualUserSoc** < 99.5%	Signal from battery to stop the charging process, due to full battery or error.	Look for other battery errors, if none read errormemory with Tool32
Battery:BMWix	0xB004	CoolingValveShortToGround	**BMWix:CoolingValveErrorState** == ShortToGround	Battery detected short circuit to GND on cooling valve near data plug.	Check and/or replace cooling valve
Battery:BMWix	0xB005	CoolingValveShortToUbat	**BMWix:CoolingValveErrorState** == ShortToUbat	Battery detected short circuit to Ubat on cooling valve near data plug.	Check and/or replace cooling valve
Battery:BMWix	0xB006	CoolingValveWireBroken	**BMWix:CoolingValveErrorState** == WireBreak	Battery detected open wire on cooling valve near data plug.	Check plug from Dataplug to cooling valve is inserted in cooling valve. Check and/or replace cooling valve
Battery:BMWix	0xB007	CoolingValveDriverError	**BMWix:CoolingValveErrorState** == DriverError	Battery detected driver fault in driver which drives cooling valve coil	Read errormemory with Tool32
Battery:BMWix	0xB008	CoolingValveValveStuckAtClosed	**BMWix:CoolingValveErrorState** == ValueStuckAtClose	Battery detected cooling valve stuck at closed position	Check and/or replace cooling valve
Battery:BMWix	0xB009	EmergencyModeActive	**BMWix:StateEmergencyMode** == Active and **BMWix:RequestService** == Active	Battery detected increased self-discharge of a cell. Max SOC is 40% technical SOC to reduce energy content of battery.	Perform Battery Service and replace effected cell module
Battery:BMWix	0xB00A	CoolingValveValveStuckAtOpen	**BMWix:CoolingValveErrorState** == ValueStuckAtOpen	Battery detected cooling valve stuck at open position	Check and/or replace cooling valve
Battery:BMWix	0xC001	Category6Failure	**BMWix:RequestOpenContactorFast** == Active	Category 6 Failure in battery (open contactors within 1-2s)	Check for other battery errors, read errormemory with Tool32
Battery:BMWix	0xC002	Category7Failure	**BMWix:RequestOpenContactorNow** == Active	Category 7 Failure in battery (open contactors immediately)	Check for other battery errors, check HVIL Plug on HV connector, read errormemory with Tool32
Battery:BMWix	0xD001	ThermalEventHazard	CAN message ID:0x587 (Check control message) received or **BMWix:StatusWarnOverTemp** == Active	Thermal runaway of a cell detected	Switch off system, distance oneself from the system
Battery:BMWix	0xFF00	DeviceSoftwareGenericFail	Unplausible Codepath taken.	Software Bug	Inform R&D